Topological Groups

A group G is a non-empty set with a map
$m: G \times G \to G$ (m for product)
(i) that is associative:

$$m(g_1, m(g_2, g_3)) = m(m(g_1, g_2), g_3) \text{ for all } g_1, g_2, g_3 \in G;$$

(ii) possesses an identity element e s/t

$$m(e, g) = m(g, e) = g \text{ for all } g \in G;$$

(iii) for each $g \in G$ there is $h \in G$ s/t

$$m(g, h) = m(h, g) = e.$$

Typically one writes $m(g,h) = gh$, with which (i) and (ii) read $g_1(g_2 g_3) = (g_1 g_2) g_3$ and $eg = ge = g$. The element h in (iii) is the inverse of g and one writes g^{-1} for it.

Note: When describing a gp, one needs to specify the group operation.

Ex: $(\mathbb{Z}, +)$
Ex: $(\mathbb{R}^+, \cdot)$ ($\mathbb{R}^+$ positive real numbers)
Ex: $(\mathbb{C}^*, \cdot)$ ($\mathbb{C}^*$ non-zero cx numbers)
Ex: $(\mathbb{R}^n, +)$ (vector addition on $\mathbb{R}^n$)
Ex: $(GL(n), \cdot)$ ($n \times n$ invertible matrices)

A topological group G is a group equipped with a topology in which the group operations $m: G \times G \to G$ and $i: G \to G$ $(i(g) = g^{-1})$ are continuous. Call i the inversion.

Ex: $(\mathbb{Z}, +)$ (with the discrete topology on $\mathbb{Z}$)
Ex: $(\mathbb{R}^+, \cdot)$
Ex: $S^1 = \{ z \in \mathbb{C} \mid |z| = 1 \}$ with cx mult.
Ex: One can identify $n \times n$ matrices with $\mathbb{R}^{n^2}$ (simply string together the rows in a matrix). This defines a topology on $M_{n \times n}$ ($n \times n$ matrices).
Let $Gl(n) \subset M_{n \times n}$ consist of all invertible matrices. Then $Gl(n)$ is an open subset ($A \in Gl(n) \iff \det A \neq 0$).

Now matrix mult is a polynomial fnc in the components and the matrix inverse is a rational function of the same. Thus $Gl(n)$ is a topological gp.

Let G be a gp. A subset $H \subset G$ is called a subgroup of G if (i) $e \in H$, (ii) $h_1 h_2 \in H$ for all $h_1, h_2 \in H$, (iii) $h^{-1} \in H$ for all $h \in H$.

Two-step subgroup test: Let $H \neq \emptyset$. If $h_1 h_2^{-1} \in H$ for all $h_1, h_2 \in H$, then $H < G$ is a subgroup.

Prop: Let G be a top. gp, and let $H < G$ be an algebraic subgroup. Equip H with the subspace topology. Then H is also a top. gp.

Sketch of proof: Let $j: H \to G$ be the inclusion map, which is cont. Thus $m \circ (j \times j): H \times H \to G$ is cont. But the image of this map lies in H, so it is cont. as a map into H. Thus $m: H \times H \to H$ is cont.
One similarly shows that $i: H \to H$ is cont.

Ex: $SL(n) = \{ A \in M_{n \times n} \mid \det A = 1 \} < Gl(n)$ ($<$ for subgp) so $SL(n)$ is a top. gp.

Ex: $SO(n) = \{ A \in M_{n \times n} \mid AA^T = I, \det A = 1 \} < Gl(n)$ is a top. gp.

Fix $h \in G$, and let $L_h: G \to G$, $R_h: G \to G$ be given by $L_h g = hg$, $R_h g = gh$ (these are the left and right translations by h). Then both are homeom of G (check!).

Thus, if V is a nbhd of e, then $L_h V$ is a nbhd of h, and, similarly, a nbhd of h gives rise to a nbhd of e.

Prop. Let G be a connected topological gp, and let V be a nbhd of e. Then V generates G.

(Meaning: Given $g \in G$, then there are $g_1, \ldots, g_k \in V$, not necessarily unique and with k depending on g, s/t $g = g_1 g_2 \cdots g_k$.)

Sketch of proof: Let $V^{-1} = \{g^{-1} \mid g \in V\}$. Since the inversion is a homeom., V^{-1} is also a nbhd of e, and then so is $W = V \cap V^{-1}$. Note that $W^{-1} = W$.

Goal: Show that the set P of finite products of elements of W forms an open and closed set.

First assume that $g \in P$. Then gW is a nbhd of g and clearly $gW \subset P$. So P is open.

Next let $g \in \neg P$. We claim that $gW \subset \neg P$. Suppose $h \in gW \cap P$. Then $h = gg_1$ for some $g_1 \in W$, so $g = hg_1^{-1}$. But $g_1^{-1} \in W$, so this would imply that $g \in P$, a contradiction. Thus $\neg P$ is open.